Lakshay Chauhan

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Education

Indraprastha Institute of Information Technology (IIIT) Delhi Jul 2021 – Jun 2025
Bachelor of Technology in Computer Science and Engineering *Delhi, India*

Work Experience

Member of Technical Staff Jul 2025 – Present
ZL Technologies, Inc | ***Website*** *Hyderabad, India*

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

Research Assistant | ***Gitlab Project Organization*** May 2024 – May 2025
Networks and Systems Security Lab - IIITD | ***Website*** *Delhi, India*

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

Projects

Minimal Compiler Infrastructure | ***Repository*** May 2024 – Present

- ***Key Skills:*** *Compiler Infrastructure, Design and Optimization, Rust, FASM, WASM, Graphviz*
- Developed a minimal compiler infrastructure inspired by LLVM. It converts source code into an optimized control flow graph (CFG) and translates it into target-specific assembly code.
- Implemented CFG optimization passes, such as identifier validation, constant folding, and CFG simplification, ensuring high-performance code generation.

Exploring Compiler Optimizations | ***Repository*** Nov 2024 – Feb 2025

- ***Key Skills:*** *Compiler Optimization, Compiler Passes, Garbage Collectors, Type Safety, LLVM, C, C++*
- Developed LLVM-based compiler and runtime systems including a null pointer dereference detection pass, a conservative bump-pointer garbage collector for C with root scanning, and memory safety mechanisms enforcing spatial and weak type safety through bounds tracking and invalid pointer prevention.

Garbage Collector for C | ***Repository*** Feb 2025 – Present

- ***Key Skills:*** *Runtime Systems, C, Garbage Collection, Memory Management, Systems Programming*
- Developed a conservative mark-and-sweep garbage collector for C with an 8-byte aligned bump allocator and segmented heap design.
- Implemented stack, heap, and global-root scanning for automatic memory reclamation.
- Designed page-level memory management using `mmap`, `mprotect`, and `madvise`.

Open Source Contribution

Active open-source contributor to **Microsoft** (primarily MSSQL integrations including `mssql-rs` and `mssql-django`), **Cloudflare**, **LLVM**, **Nushell**, **GitHub**, and other projects, contributing issues, features, bug fixes, developer tooling, pull requests, code reviews, and documentation. A complete timeline of contributions is available at **osc.9th.fun**.

Technical Skills

Languages: C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell
Frameworks: PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib
Developer Tools: Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SqlLite3
Technical Electives: Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing